

Alex Road Service

Website + Operations Platform

Technical Product Specification

Project: Alex Road Service Digital Platform

Developer: Zakiy Manigo

Client: Alex Road Service – Commercial Truck & Trailer Repair

Location: Woodbridge, New Jersey

1. Project Overview

This project will create a **single web platform** consisting of:

1. **Public Marketing Website**
2. **Internal Operations Platform**

The public website will allow customers to learn about Alex Road Service and request service.

The operations platform will allow the business to manage:

- customers
- fleets
- trucks
- work orders
- estimates
- invoices
- payments
- inventory
- service history

Both systems will live under **one website domain**.

Example structure:

alexroadservice.com

Public pages:

alexroadservice.com

Operations platform login:

alexroadservice.com/login

Dashboard:

alexroadservice.com/app

2. Technology Stack

Frontend

React / Next.js

Responsive UI (desktop, tablet, mobile)

Backend

Firebase

Services used:

- Firebase Authentication
- Firestore Database
- Firebase Storage
- Firebase Hosting
- Firebase Functions

Google Cloud

- Google Maps API
- Google Analytics
- Google Search Console
- Google Cloud Storage

Additional Tools

3. Product Architecture

Public Website

| Login



Operations Platform (Web App)

| Customers
| Fleet Records
| Work Orders
| Estimates
| Invoices
| Payments
| Inventory
| Service History
| Reports

4. Product Components

4.1 Marketing / Business Website

Purpose:

- Generate leads
- Show services
- Improve local search presence
- Allow service requests
- Provide login access to the operations system

Pages

Home

- hero section

- services overview
- call button
- request service button
- reviews/testimonials

About

- company story
- certifications
- experience

Services

- commercial truck repair
- trailer repair
- roadside assistance
- fleet maintenance

Emergency Roadside Page

- emergency contact
- service request form
- map location

Contact

- contact form
- phone number
- address
- Google Maps integration

Reviews

- customer testimonials
- Google review integration

Financing / Payment Page

- payment request
- invoice payment link

SEO Requirements

Local search optimization for:

truck repair woodbridge nj
commercial truck repair nj
mobile truck repair nj
trailer repair nj
fleet truck service nj

SEO elements:

- structured metadata
 - schema markup
 - sitemap
 - optimized page speed
 - mobile responsive design
-

5. Operations Platform (Internal Web App)

Purpose:

To replace manual workflows and allow the shop to manage daily operations digitally.

Accessed via:

alexroadservice.com/login

Users:

- Admin
 - Office staff
 - Technicians
-

6. Core Modules

6.1 Dashboard

Overview metrics:

- open work orders

- jobs in progress
 - completed jobs
 - revenue today
 - pending invoices
 - low inventory alerts
-

6.2 Customer Management

Store customer accounts.

Customer fields:

customer_id
company_name
contact_name
phone
email
address
notes
created_at

Customer profile displays:

- company info
 - truck fleet
 - service history
 - invoices
 - notes
-

6.3 Fleet & Truck Records

Each customer may have **multiple trucks**.

Truck fields:

truck_id
customer_id
unit_number
VIN

make
model
year
license_plate
mileage
notes

Truck profile includes:

- service history
- repairs
- oil change history
- maintenance reminders

6.4 Work Orders

Work orders represent active jobs.

Fields:

work_order_id
customer_id
truck_id
technician
description
status
labor_hours
labor_rate
parts_used
created_at
completed_at

Statuses:

- Open
 - In Progress
 - Waiting Parts
 - Completed
 - Invoiced
-

6.5 Estimates

Create repair estimates before work begins.

Fields:

estimate_id
customer_id
truck_id
parts
labor_hours
labor_rate
tax
total
status

Status:

- Draft
- Sent
- Approved
- Rejected

Estimates can convert directly into invoices.

6.6 Invoices

Invoices are generated from work orders or estimates.

Fields:

invoice_id
customer_id
work_order_id
parts_total
labor_total
tax
total
status
payment_status

Invoice actions:

- generate PDF
 - email invoice
 - generate payment link
-

6.7 Online Payments

Integrations:

- Stripe

Features:

- payment link
 - credit card payments
 - ACH payments
 - invoice payment tracking
-

6.8 Parts Inventory

Track parts used in repairs.

Fields:

part_id
name
category
cost
sale_price
quantity
supplier
low_stock_threshold

Features:

- stock tracking
- parts usage per job
- low inventory alerts

6.9 Service History

Each truck will have a full history.

Includes:

- repairs
- oil changes
- maintenance
- technician notes
- photos

6.10 Maintenance Reminders

Automated reminders for:

- oil changes
- maintenance intervals

Triggers based on:

- time
- mileage

6.11 Technician Workflow

Technicians can:

- view assigned jobs
- update job status
- log labor hours
- upload repair photos
- add notes

Mobile-friendly UI.

6.12 Reporting Dashboard

Reports include:

- revenue by month
 - jobs completed
 - labor hours
 - top customers
 - parts usage
-

7. Integrations

Payments

Stripe

Accounting

QuickBooks (future phase)

Communication

Twilio (future SMS updates)

Email

Send invoices via email

Maps

Google Maps API

Storage

Firebase Storage for:

- repair photos

- documents
 - invoices
-

8. Mobile Support

The operations platform will be **mobile responsive**, allowing technicians to use it from:

- phones
- tablets
- laptops

Native mobile apps are not required for Phase 1.

9. Phase 1 Deliverables

Website

- public marketing site
- SEO optimized
- mobile responsive
- contact forms
- login access

Operations Platform

- secure login
 - dashboard
 - customers
 - fleet records
 - work orders
 - estimates
 - invoices
 - payments
 - inventory
 - service history
 - reports
-

10. Future Phase

Upgrades:

- QuickBooks sync
 - SMS repair updates
 - GPS roadside dispatch
 - technician scheduling
 - AI maintenance predictions
 - fleet analytics
-

11. Security

Security features:

- Firebase authentication
 - role-based access
 - encrypted database
 - secure payment processing
 - HTTPS hosting
-

12. Hosting

Hosted using:

- Firebase Hosting
 - Firestore database
 - Firebase storage
 - Google Cloud infrastructure
-